

Predicting Occupational Gender-Typing from Job-Advertisement Language: A Machine-Learning Extension of a Gendered-Wording Study

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Abstract

A prior study of this dataset found that job advertisements in male-dominated occupations use slightly more agentic (masculine-coded) wording than those in female-dominated occupations, a small effect (Cohen’s $d = 0.24$), while communal wording did not differ. We extend that finding to a prediction task: can the language of a single advertisement predict whether it belongs to a male-dominated or a female-dominated occupation? Using 4,297 real advertisements from the Employment Scam Aegean Dataset, we compare a majority-class baseline against two logistic-regression models on a held-out test set, with all preprocessing fit inside training-only pipelines. A model using only the gendered-wording features barely separates the two groups (ROC-AUC = 0.60) and does not beat the baseline on accuracy, exactly as the small inferential effect predicts. A full-text model reaches ROC-AUC = 0.99, but its most predictive terms are occupation nouns (*engineer*, *software* versus *patient*, *teaching*), showing that it detects the job, not its gendered tone. Gendered wording is thus a real but subtle aggregate signal that does not classify individual advertisements.

1 Introduction

Recruitment language may subtly shape who applies for a job. In earlier work on this dataset we tested the claim of Gaucher, Friesen, and Kay. that male-dominated occupations advertise with more *agentic* (masculine-coded) wording and female-dominated occupations with more *communal* (feminine-coded) wording. We found a statistically significant but small agentic difference (Cohen’s $d = 0.24$) and no communal difference: the gendered signal was real but subtle, and asymmetric. That analysis operated at the group level, comparing average wording between occupation groups.

This paper asks a complementary, individual-level question that requires machine learning: given the language of a *single* advertisement, can we predict whether it comes from a male-dominated or a female-dominated occupation? First, from a practical standpoint if gendered wording were a strong signal, an ad’s language alone would reveal the gender-typing of its field, and tools could flag or rewrite ads to broaden the applicant pool. If it is a weak signal, then wording is only a subtle cue and interventions based on it should be modest in their claims. Second, the project also provides an honest stress-test of the earlier result a small average difference should translate into weak predictive power, and confirming that keeps our conclusions calibrated. To separate genuine gendered-wording signal from mere topic vocabulary, we compare two feature sets: the gendered-wording features alone, and the full advertisement text. Both are measured against a majority-class baseline.

2 Method

2.1 Data and sample

We reuse the Employment Scam Aegean Dataset (EMSCAD), obtained from its public Kaggle distribution. As in the earlier study, we kept only legitimate postings carrying an occupation (*function*) label and a description, dropped advertisements shorter than 30 word tokens, and restricted attention to six functions grouped by workforce gender composition from U.S. Bureau of Labor Statistics data: engineering and information technology (male-dominated; about 17% and 26% women) versus administrative support, health care, human resources, and education (female-dominated; roughly 73–76% women). This yielded 4,297 advertisements, of which 2,933 (68.3%) were male-dominated and 1,364 female-dominated.

2.2 Prediction task and features

The unit of analysis is one advertisement and the target is binary (male-dominated = 1). We build two feature sets. The *wording* set is the four features from the earlier study: agentic and communal words per 100 words, their difference, and advertisement length. The *text* set is a TF-IDF representation of the full advertisement (title, description, and requirements), limited to the 5,000 most frequent terms with English stop-words removed. The wording features test the gendered-language hypothesis directly; the text features probe how much predictive power the raw language contains and where it comes from.

2.3 Models, evaluation, and leakage control

We held out 25% of advertisements as a stratified test set and evaluated every model only on it. Preprocessing (standardization for the wording features, TF-IDF fitting for the text) was placed inside scikit-learn pipelines so that it is fit on training data alone, preventing leakage; the same guarantee holds within each cross-validation fold. The *baseline* always predicts the majority class. *Model A* is logistic regression on the wording features with balanced class weights, so that it attends to the minority class rather than collapsing onto the majority. *Model B* is logistic regression on the TF-IDF text. Logistic regression was chosen because it suits binary text classification and yields interpretable coefficients. Because the classes are imbalanced, we report ROC-AUC and balanced accuracy alongside accuracy, precision, recall, and F_1 , and we confirm the held-out numbers with 5-fold stratified cross-validation. We interpret models through their coefficients and check overfitting by comparing training and test performance. No causal claims are drawn from predictive accuracy.

Table 1: Held-out test performance ($n = 1075$). Acc: accuracy; BAcc: balanced accuracy; P/R: precision/recall; AUC: ROC-AUC.

Model	Acc	BAcc	P	R	F_1	AUC
Baseline (majority)	.683	.500	.683	1.00	.811	.500
A: gendered wording	.561	.568	.741	.549	.631	.604
B: full ad text	.975	.964	.969	.995	.982	.994

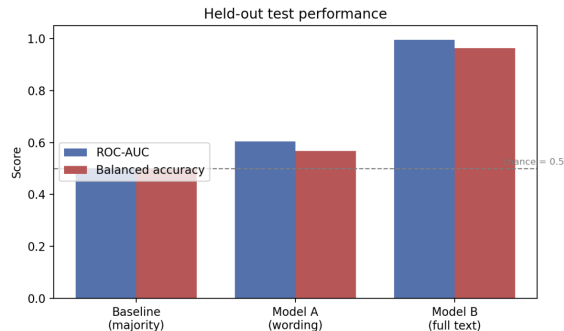


Figure 1: ROC-AUC and balanced accuracy on the held-out test set. The gendered-wording model (A) sits just above chance; the full-text model (B) is near-perfect. Dashed line marks chance (0.5).

3 Results

3.1 Model comparison

Table 1 and Figure 1 report held-out performance. The baseline reaches 68.3% accuracy by always predicting the majority class, with a balanced accuracy and ROC-AUC of 0.50 by construction. Model A, using gendered wording alone, achieves a ROC-AUC of only 0.60 and a balanced accuracy of 0.57; its raw accuracy (0.561) is *below* the baseline, because forcing it to attend to both classes trades majority-class accuracy for minority recall. Five-fold cross-validation agrees ($AUC = 0.61 \pm 0.02$). Model B, using the full text, reaches 0.975 accuracy and a ROC-AUC of 0.99 (cross-validated 0.996 ± 0.001).

3.2 What the text model uses

Model B’s accuracy does not imply that gendered wording is highly predictive. Figure 2 shows its most influential terms: predictions toward male-dominated occupations are driven by *engineer*, *software*, *developer*, and *systems*, and toward female-dominated occupations by *patient*, *teaching*, *administrative*, and *care*. These are occupation nouns, not agentic or communal tone words. The full-text model is therefore recovering the job itself from its topical vocabulary.

3.3 Overfitting

Model A shows essentially identical training and test balanced accuracy (0.562 vs. 0.568); it is too simple to overfit. Model B’s training accuracy (0.987) is only slightly above its test accuracy

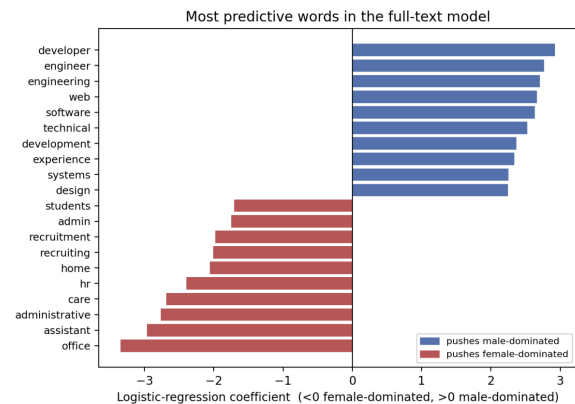


Figure 2: Most predictive terms in the full-text model (B). The strongest features are occupation nouns rather than gendered tone words.

(0.975), and cross-validation matches the held-out figure, so it is not badly overfit either.

4 Discussion

The two models together sharpen the earlier inferential finding. Gendered wording carries a genuine but weak signal: Model A ranks advertisements only slightly better than chance ($AUC = 0.60$) and cannot beat a majority-class guess. This is the individual-level counterpart of the small group-level effect ($d = 0.24$) reported earlier, a difference detectable in averages over thousands of ads but far too small to classify any single one. The near-perfect full-text model does not contradict this; it succeeds by reading occupation vocabulary, which is a proxy for the label rather than a measure of gendered tone. What must *not* be inferred from Model B is that language strongly encodes a field’s gender-typing through gendered wording specifically.

Several limitations apply. The occupation groups are coarse and are labelled from external workforce statistics, not from applicant gender, which the corpus does not record; the advertisements come from one platform and the 2012–2014 period; and the lexicon counts words out of context. Ethically, a classifier that infers a field’s gender-typing from text could reinforce stereotypes if used to screen or route advertisements, and predictive accuracy is not evidence about who actually applies or why. We therefore make no causal claims.

5 Conclusion

Extending a gendered-wording study into a prediction task, we have found that gendered wording alone predicts an advertisement’s occupational gender-typing only marginally better than chance, whereas the full text predicts it almost perfectly: by recognizing the occupation, not its gendered tone. The contribution is to locate the gendered-wording effect precisely: a subtle aggregate signal, not an individual-level discriminator. Methodologically, the study illustrates baseline-anchored, leakage-controlled evaluation and

the importance of interpreting *what* a high-accuracy text model has actually learned.

References

- [1] Danielle Gaucher, Justin Friesen, and Aaron C. Kay. 2011. Evidence that gendered wording in job advertisements exists and sustains gender inequality. *Journal of Personality and Social Psychology* 101, 1 (2011), 109–128. <https://doi.org/10.1037/a0022530>
- [2] Mihwa Seong and Simon C. Parker. 2024. Does gendered wording in job advertisements deter women from joining start-ups? A replication and extension of Gaucher, Friesen, and Kay (2011). *Strategic Entrepreneurship Journal* 18, 2 (2024), 286–305. <https://doi.org/10.1002/sej.1489>
- [3] Sokratis Vidros, Constantinos Koliass, Georgios Kambourakis, and Leman Akoglu. 2017. Automatic detection of online recruitment frauds: Characteristics, methods, and a public dataset. *Future Internet* 9, 1 (2017), 6. <https://doi.org/10.3390/fi9010006>
- [4] Shivam Bansal. 2020. Real / Fake Job Posting Prediction [Data set]. Kaggle. <https://www.kaggle.com/datasets/shivamb/real-or-fake-fake-jobposting-prediction/data>
- [5] U.S. Bureau of Labor Statistics. 2024. Labor Force Statistics from the Current Population Survey, Table 11: Employed persons by detailed occupation, sex, race, and Hispanic or Latino ethnicity. <https://www.bls.gov/cps/cpsaat11.htm>